

Dark Matter Effect

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A field-based interpretation of the dark matter effect as a consequence of non-coupling-capable structures in a continuous medium

Abstract

We present a systematic analysis of 163 galactic rotation curves from the SPARC database, focusing on the empirical additional term $C(r) = v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r)$. After normalization $F(x) = C(r)/C_{\text{max}}$ with $x = r/R$, global fits across subsamples reveal a consistent structure of preferred regimes, while complementary individual fits yield a broader but still clearly structured distribution of the scale parameter q .

The global fits identify two preferred dynamical regimes:

Peak regime: characterized by a pronounced radial peak with scale parameter $q \approx 0.5$ – 1.0 . This regime includes, in particular, low-surface-brightness galaxies (F-type) as well as selected dwarf galaxies.

Diffuse regime: characterized by a broadly distributed profile with $q \approx 3.0$. This regime is dominated primarily by more massive spiral galaxies.

An additional analysis of individual fits across 164 successfully fitted systems shows that approximately 62% of galaxies lie in the peak regime, about 26% in the diffuse regime, and about 12% in the transition region $1.2 \leq q < 2.5$. The distribution thus exhibits two preferred dynamical states with a weakly but significantly populated transition zone.

In addition, a systematic dependence on the baryonic scaling parameter a is observed: at low values ($a \approx 0.6$), the peak structure remains stable, whereas at high values ($a \approx 0.8$), a shift toward more diffuse profile shapes sets in.

The concentration in preferred parameter ranges is not the natural expectation of continuous halo models, which would predict a smoother distribution of profile shapes. Instead, the results point to preferred dynamical states that are systematically coupled to the baryonic structure.

Within the framework of UQSH, the dark matter effect is interpreted as the consequence of persistent, not fully coupling-capable field organization generated by baryonic structures acting both as a source of continuous field excitation and as an anchor of its large-scale stabilization.

The universal fits achieve mean squared errors of $\text{MSE} \approx 0.016$ in the peak regime and $\text{MSE} \approx 0.06\text{--}0.13$ in the diffuse regime.

Contents

1	Introduction	5
1.1	Introduction	5
1.2	Intuitive classification of the additional dynamics	5
1.3	Motivation and objectives	6
1.4	Structure of the paper	6
2	Observation-based motivation	7
2.1	Galactic rotation curves	7
2.2	Radial Acceleration Relation	7
2.3	Field-based interpretation	8
2.4	Effective density model	8
3	Data and methodology	8
3.1	SPARC database	8
3.2	Extraction of the additional term	9
3.3	Normalization and shape function	10
3.4	Parametric fit	10
3.5	Classification of the regimes	10
4	Results	11
4.1	Structure of the additional term	11
4.2	Bimodal regime structure	12
4.2.1	Peak regime	12
4.2.2	Diffuse regime	13
4.3	Occupancy of the transition region	13
4.4	Dependence on the baryonic scaling parameter	16
4.5	Universality within the regimes	17
4.6	Complete description of the rotation curves	17
5	Physical interpretation	18
5.1	Field-based interpretation of the regime structure	18
5.1.1	Peak regime: localized reorganization	18
5.1.2	Diffuse regime: large-scale field response	19
5.2	Conceptual classification of the dark matter effect	19
5.2.1	Persistent field organization as the origin of the dark matter effect	20

5.2.2	Baryonic structure as source and anchor	20
5.2.3	Non-coupling-capable field organization and observability	21
5.3	Interpretation within the framework of UQSH	21
5.3.1	Coupling and threshold crossing	21
5.3.2	Scale analogy of localized coupling modes	22
5.4	Interpretation of the transition region	22
5.4.1	Preferred field modes	22
5.4.2	Dynamical transition	23
5.5	Coupling to baryonic structure	23
5.6	Testable predictions	24
5.6.1	Galaxy evolution and regime transition	24
5.6.2	Spatial distribution of subsystems	24
5.6.3	Temporal evolution	26
6	Comparison with standard models	26
6.1	Expectations from the Λ CDM model	26
6.2	Observed discrepancy	26
6.3	Possible explanation within the Λ CDM framework	27
6.4	Distinction between the approaches	27
7	Discussion and outlook	28
7.1	Main results	28
7.2	Interpretation and significance	28
7.3	Open questions and future work	29
7.3.1	Physical derivation of the shape function	29
7.3.2	Scaling relations	29
7.3.3	Spatial correlations	30
7.3.4	Extension to other observables	30
7.3.5	Comparison with simulations	30
7.4	Significance for the understanding of dark matter	31
8	Conclusion	31

1 Introduction

1.1 Introduction

Observations of galactic rotation curves, gravitational lensing, and large-scale structure formation indicate that the gravitational effect of visible matter is not sufficient to explain the dynamics of astrophysical systems. This discrepancy is usually addressed by introducing dark matter, modeled as an additional non-luminous particle component.

Despite its success at the phenomenological level, the particle nature of dark matter has so far not been confirmed experimentally. This has led to the development of alternative approaches that reinterpret the origin of the observed effects.

In this work, a field-based approach is pursued in which physical reality is described by a continuous field medium. Observable matter corresponds to stable, coupling-capable organizational states of this field. It is proposed that additional non-coupling-capable structures exist that contribute to gravitational dynamics without being directly observable.

Instead of introducing new particles, dark matter is thus interpreted as an effect of field organization.

1.2 Intuitive classification of the additional dynamics

For an intuitive interpretation, the empirical additional term

$$C(r) = v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r)$$

can be understood as an additional field-mechanical response that appears beyond the baryonic contribution.

In simple terms, $C(r)$ describes how strongly the underlying field medium responds additionally to the existing matter structure.

The analysis shows that this additional effect is not distributed arbitrarily, but instead assumes two preferred forms:

- **Localized response (peak regime):** the additional effect is concentrated within a specific radial region and exhibits a pronounced maximum.

- **Distributed response (diffuse regime):** the additional effect is broadly distributed over large radii and does not exhibit a characteristic maximum.

In intuitive terms, this corresponds to two different kinds of field response: either a locally concentrated enhancement or a large-scale, uniformly distributed response.

1.3 Motivation and objectives

The present work pursues two central aims:

1. **Empirical analysis:** systematic extraction and characterization of the effective additional term $C(r)$ from a large sample of rotation curves from the SPARC database.
2. **Physical interpretation:** investigation of whether the observed structure of the additional term is consistent with a field-based interpretation within the framework of the Universal Quantum Foam Hypothesis (UQSH).

1.4 Structure of the paper

The paper is structured as follows:

Section 2 provides an overview of the observational motivation for dark matter and introduces the basic field-based interpretation.

Section 3 describes the data used and the methodology for extracting the additional term.

Section 4 presents the central empirical results, in particular the identification of preferred dynamical regimes.

Section 5 develops the physical interpretation within the framework of the Universal Quantum Foam Hypothesis (UQSH), including a conceptual classification of the dark matter effect as an expression of persistent field organization and the resulting testable predictions.

Section 6 compares the results with standard models and discusses the significance of the observed regime structure.

Section 7 summarizes the main results and provides an outlook on future work.

2 Observation-based motivation

2.1 Galactic rotation curves

A central motivation for dark matter arises from galactic rotation curves. For a particle at distance r from the center, classical mechanics yields

$$v(r) \sim \sqrt{\frac{GM(r)}{r}},$$

which, for a bounded mass distribution, would imply a decline in velocity.

Observations, however, show that rotation curves in the outer regions of galaxies are typically flat:

$$v(r) \approx \text{constant}.$$

This implies that the effective enclosed mass $M(r)$ continues to increase with radius.

Additional evidence arises from gravitational lensing and from the dynamics of galaxy clusters, which likewise point to the presence of additional gravitational contributions that are not electromagnetically visible.

2.2 Radial Acceleration Relation

A particularly important empirical indication of the close coupling between visible matter and additional dynamics is the so-called Radial Acceleration Relation (RAR) [1].

This relation describes the observed connection between the radial acceleration expected from the baryonic matter distribution and the dynamical acceleration actually inferred from the rotation curves. The additional gravitational effect does not appear as a freely distributed component independent of the baryonic structure, but instead follows the visible matter distribution in a strikingly systematic way.

Within particle-based halo models, this close coupling constitutes an additional structural feature that must be explained. Within the field-based interpretive framework of the present work, by contrast, it can be read as a natural indication that the additional dynamics does not arise independently of baryonic organization, but is closely linked to it.

2.3 Field-based interpretation

We consider a continuous field medium whose organization exhibits different stability and coupling states.

A distinction is made between:

- coupling-capable field structures, which appear as visible matter,
- non-coupling-capable field structures, which are not directly observable.

The central assumption is that both types of structure contribute to gravitational dynamics. Non-coupling-capable field structures are not to be understood as physically ineffective, but merely as not directly visible within the respective observational regime. Observable matter therefore represents only part of the total field organization.

2.4 Effective density model

To describe the gravitational effect, we introduce an effective mass density:

$$\rho_{\text{eff}} = \rho_{\text{visible}} + \rho_{\text{sub}},$$

where ρ_{sub} describes the contributions of non-coupling-capable field structures.

The gravitational potential equation then becomes

$$\nabla^2 \Phi = 4\pi G \rho_{\text{eff}}.$$

For extended distributions of ρ_{sub} , the effective mass can continue to increase with radius, leading to flat rotation curves.

This provides a natural explanation for observed galactic dynamics without introducing additional particle components.

3 Data and methodology

3.1 SPARC database

The present analysis is based on rotation curves from the SPARC database (Spitzer Photometry and Accurate Rotation Curves) [2]. SPARC includes high-quality rotation curves of 175 galaxies with:

- HI and/or H α velocity measurements
- photometric data at 3.6 μm (Spitzer)
- mass models for gas, disk, and bulge
- radii up to 20–40 kpc
- mass range: 10^8 to $10^{11} M_\odot$

Of the available 175 galaxies, 163 systems were included in the present analysis after excluding systems with incomplete data or insufficient radial coverage.

3.2 Extraction of the additional term

For each galaxy, the effective additional term was determined directly from the observational data. The starting point is the relation

$$v_{\text{obs}}^2(r) = v_{\text{bar}}^2(r) + C(r),$$

where $v_{\text{obs}}(r)$ is the observed rotation velocity and $v_{\text{bar}}(r)$ is the baryonic contribution.

The baryonic contribution is composed of:

$$v_{\text{bar}}^2(r) = v_{\text{gas}}^2(r) + a v_{\text{disk}}^2(r) + b v_{\text{bulge}}^2(r),$$

where a and b are scaling parameters. In the standard analysis, $b = 1$ was fixed and a was systematically varied over the range $a \in [0.2, 1.0]$.

The additional term thus follows directly as

$$C(r) = v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r).$$

Negative values of $C(r)$ were interpreted as non-physical and set to zero. This typically affects the outermost radial region, where measurement uncertainties may dominate.

3.3 Normalization and shape function

To make different galaxies comparable, a normalization was introduced:

$$x = \frac{r}{R}, \quad F(x) = \frac{C(r)}{C_{\max}},$$

where $C_{\max}$ is the maximum of $C(r)$ and R is the radial position of this maximum.

The normalized function $F(x)$ describes only the shape of the additional term, independent of amplitude and physical scaling.

3.4 Parametric fit

For the normalized shape function, a parametric ansatz was chosen:

$$F(x) = \frac{x^p e^{-x/q}}{\max_{y>0} (y^p e^{-y/q})},$$

where p describes the shape exponent and q the scale parameter.

Normalization by the maximum ensures that $\max F(x) = 1$.

The parameters p and q were determined individually for each galaxy by minimizing the quadratic deviation:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N [F(x_i) - F_{\text{fit}}(x_i)]^2.$$

In addition, global fits over subsamples were carried out to test whether groups of galaxies share a common shape function.

3.5 Classification of the regimes

Based on the extracted parameters, the galaxies were classified into two main regimes:

- **Peak regime:** $q < 1.2$ and a pronounced internal maximum of $F(x)$
- **Diffuse regime:** $q \geq 2.5$ and a monotonic or broadly distributed profile

The transition region $1.2 \leq q < 2.5$ was examined separately in order to quantify its occupancy.

The chosen boundaries reflect the observed clustering of parameter values and serve to clearly separate the dominant regions in parameter space. The underlying distribution is continuous, but shows clearly preferred regions.

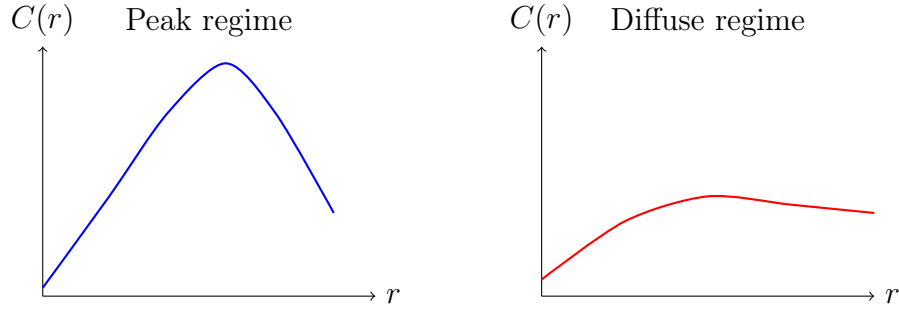


Figure 1: Schematic representation of the two dynamical regimes of the additional term $C(r)$. In the peak regime, the additional field effect is localized and exhibits a pronounced maximum. In the diffuse regime, the field effect is broadly distributed without a characteristic peak.

4 Results

4.1 Structure of the additional term

The functions $C(r)$ extracted from the observational data show consistent qualitative features across different galaxies:

- an initial rise in the inner region,
- a pronounced maximum at a characteristic scale R ,
- followed by a flattening or slight decline in the outer region.

This structure differs significantly from classical monotonic halo profiles (e.g., NFW) and points to a non-trivial radial dynamics.

4.2 Bimodal regime structure

After normalization of the additional terms according to $F(x) = C(r)/C_{\max}$ and $x = r/R$, global fits across subsamples reveal a consistent structure of preferred regimes.

Table 1 summarizes these global results for different galaxy subsamples.

Table 1: Overview of global fits across different galaxy subsamples. Listed are the number of analyzed systems (n), typical fit parameters (p , q), and the associated dominant regime structure.

Sample	n	p	q	Regime
F galaxies	15–16	1.2–1.6	0.7–1.0	Peak-dominated
D galaxies	4	1.0–1.4	0.5–0.8	Peak-dominated
ESO / DDO	9	0.66–0.70	~ 3.0	Diffuse-dominated
IC / KK	3	0.8–1.1	~ 3.0	Diffuse-dominated
NGC	58–63	0.62–0.93	~ 3.0	Diffuse-dominated
UGC / UGCA	77–79	0.66–0.78	~ 3.0	Diffuse-dominated
Total	163			

4.2.1 Peak regime

The peak regime is characterized by:

- scale parameter $q \approx 0.5\text{--}1.0$
- shape exponent $p \approx 1.0\text{--}1.6$
- pronounced internal maximum of the normalized shape function
- mean squared error $\text{MSE} \approx 0.016$

In the global fits, this regime is dominated in particular by low-surface-brightness galaxies (F-type) as well as selected dwarf galaxies.

After normalization, the profiles of these systems collapse onto a common universal curve, described by:

$$F(x) = \frac{x^{1.22}e^{-x/0.45}}{\max_{y>0}(y^{1.22}e^{-y/0.45})}.$$

4.2.2 Diffuse regime

The diffuse regime is characterized by:

- scale parameter $q \approx 3.0$
- shape exponent $p \approx 0.6\text{--}0.9$
- monotonic or broadly distributed profile without a pronounced peak
- mean squared error $\text{MSE} \approx 0.06\text{--}0.13$

In the global fits, this regime is dominated primarily by the large catalog groups NGC and UGC/UGCA as well as IC/KK systems and parts of the ESO/DDO subsamples.

The typical shape function for this regime is:

$$F(x) = \frac{x^{0.7} e^{-x/3.0}}{\max_{y>0} (y^{0.7} e^{-y/3.0})}.$$

4.3 Occupancy of the transition region

An additional analysis based on individual fits for all 164 successfully evaluated galaxies shows that the distribution of the scale parameter q exhibits two preferred regions, but no strictly empty transition zone.

Specifically, approximately 62.2% of systems lie in the peak regime ($q < 1.2$), about 25.6% in the diffuse regime ($q > 2.5$), while the region $1.2 \leq q < 2.5$ contains about 12.2% of the galaxies.

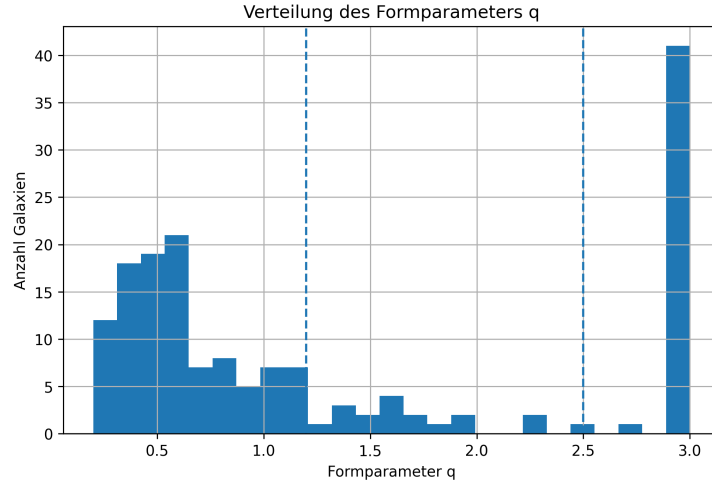


Figure 2: Distribution of the shape parameter q for all analyzed SPARC galaxies. The distribution shows a bimodal structure with a dominant population at small q values (peak systems) and a secondary concentration at large $q \approx 3$ (diffuse systems). The transition region is weakly but not empty populated, indicating a continuous transition between the dynamical regimes.

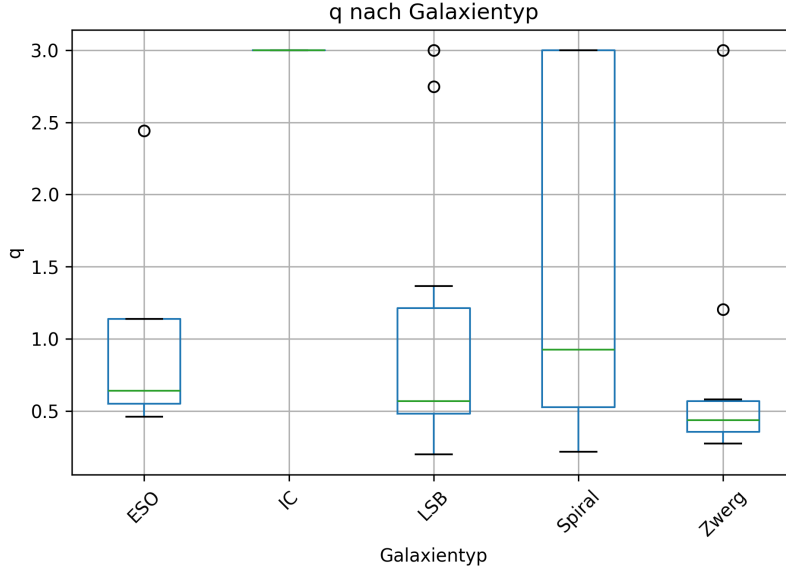


Figure 3: Distribution of the shape parameter q by galaxy type. Dwarf and low-surface-brightness galaxies show a clear concentration in the peak regime, whereas large spiral galaxies display a broader distribution.

As Figure 2 shows, the distribution of the shape parameter q is concentrated in two preferred regions: a peak-dominated region at small q values and a secondary concentration at $q \approx 3$.

The breakdown by galaxy type in Figure 3 shows that this distribution is not random, but is instead systematically associated with the respective galaxy class.

Dwarf galaxies and low-surface-brightness galaxies lie predominantly in the peak regime, whereas large spiral galaxies display a much broader distribution across peak, transition, and diffuse states.

The distribution is therefore not strictly bipartite, but instead exhibits two preferred dynamical states with a weakly populated but genuinely present transition zone.

The observed regime structure is not an artifact of the chosen parameterization. It is already visible in the normalized data themselves and remains stable under variation of the fit parameters.

Table 2: Distribution of the dynamical regimes by galaxy type based on the individual fits.

Type	Peak	Transition	Diffuse
Dwarf galaxies (DDO/D)	6	1	0
ESO	3	1	0
IC	0	0	2
LSB (F)	11	2	2
NGC	30	9	21
UGC/UGCA	51	7	17
Other	1	0	0

The occupancy of the three regions follows approximately the ratio

$$\text{Peak} : \text{Transition} : \text{Diffuse} \approx 31 : 6 : 13.$$

This shows that the peak regime dominates in the present sample, while the transition region is significantly less populated.

4.4 Dependence on the baryonic scaling parameter

The regime structure shows a systematic dependence on the baryonic scaling parameter a , which describes the weighting of the stellar disk.

For low values $a \approx 0.6$:

- peak structure remains stable
- clear separation of preferred regions in the global fits
- low scatter within the respective subsamples

For high values $a \approx 0.8$:

- transition toward more diffuse behavior even for originally peak-dominated systems
- deterioration of fit quality
- larger scatter of the parameters

This shows that the observed additional dynamics is not independent of the baryonic structure, but is systematically coupled to it.

4.5 Universality within the regimes

Within each regime, the normalized profiles show a remarkable universality:

Peak regime: after normalization, the profiles of different F and D galaxies collapse onto a common curve, with a scatter that can typically be explained by measurement uncertainties ($\text{MSE} \approx 0.016$).

Diffuse regime: here too, a consistent form is found across different catalog groups (NGC, UGC/UGCA), with the higher scatter ($\text{MSE} \approx 0.06\text{--}0.13$) attributable to the greater diversity of the included systems.

This universality stands in contrast to the expectation of individually adjusted halo profiles and points to a generic property of the underlying dynamics.

4.6 Complete description of the rotation curves

The complete description of the observed rotation curves is given by

$$v_{\text{obs}}^2(r) = v_{\text{bar}}^2(r) + C_{\text{max}} F\left(\frac{r}{R}\right),$$

where the shape $F(x)$ is regime-specific and only the amplitude C_{max} and the scale R vary from galaxy to galaxy.

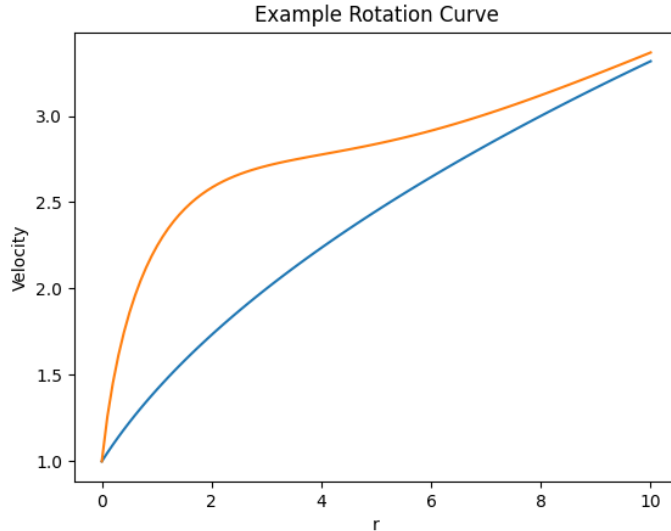


Figure 4: Schematic representation of a rotation curve. The observed velocity results from the baryonic contribution and an additional field-induced dynamics. This additional dynamics corresponds to the empirical additional term $C(r)$ and typically exhibits a non-trivial radial structure.

Typical residuals between model and observation lie in the range

$$\langle \Delta v \rangle \approx 1\text{--}3 \text{ km/s},$$

which is within the typical measurement uncertainties.

5 Physical interpretation

5.1 Field-based interpretation of the regime structure

Within the field-based interpretation proposed here, the observed regimes are understood as the expression of different organizational modes of the underlying field medium.

5.1.1 Peak regime: localized reorganization

The peak regime is interpreted as the signature of a localized reorganization dynamics of the field. In this picture:

- the baryonic structure forms an organizational core,
- it induces an additional field organization in the intermediate radial region,
- whose characteristic scale is determined by an intrinsic reorganization length ℓ_{reorg} .

The scale parameter q then corresponds approximately to the ratio:

$$q \sim \frac{\ell_{\text{reorg}}}{R_{\text{system}}}.$$

For $q \approx 0.7$, this would mean that the reorganization length is about 70% of the system size, resulting in a clearly localized structure with a pronounced peak.

5.1.2 Diffuse regime: large-scale field response

The diffuse regime is interpreted as the expression of a large-scale field response without local reorganization. In this picture:

- the field response extends over a much larger scale,
- no localization onto a characteristic reorganization length occurs,
- the effective additional dynamics is distributed over a large radial range.

The large value $q \approx 3.0$ suggests that the effective reorganization length substantially exceeds the system size, resulting in a broadly distributed, diffuse profile.

5.2 Conceptual classification of the dark matter effect

Within the framework of UQSH, the dark matter effect is not understood as the expression of an additional matter component, but rather as a consequence of the field-mechanical organization of physical space itself.

Physical space is described as a continuous, dynamical field medium whose local organizational state is characterized by a field $\phi(x, t)$. Gravitational effects arise from spatial gradients of this field state and are therefore not

restricted to bound matter, but occur in principle for any form of field excitation.

Within this framework, a distinction is made between bound field organization, which appears as stable matter structure, and free field excitation, which propagates dynamically through the medium and is phenomenologically observed as radiation.

Both do not represent separate physical entities, but rather different organizational states of the same underlying field medium.

5.2.1 Persistent field organization as the origin of the dark matter effect

Free field excitations generate real deformations of the field medium that are not strictly local and do not relax instantaneously.

In astrophysical systems, the continuous generation of such excitations leads to a permanent superposition of field deformations. A substantial fraction of these superpositions does not form stable bound states, yet remains field-mechanically effective.

The large-scale, persistent field organization arising from this manifests macroscopically as the dark matter effect.

5.2.2 Baryonic structure as source and anchor

Within this framework, baryonic structures assume a double role. On the one hand, radiatively active baryonic systems such as stars act as a continuous source of free field excitation. Radiative processes continuously generate deformations of the field medium and thus contribute directly to large-scale field organization.

On the other hand, baryonic structures constitute stable, bound organizational states of the field and thus act as local anchor regions of the field structure. They generate a spatial pre-tension of the field medium that enables and stabilizes the superposition, coherence, and persistence of field deformations.

The dark matter effect thus arises from the interplay of continuous field excitation and its structural stabilization through baryonic organization. It is therefore systematically coupled to the baryonic structure, without being caused solely by it.

5.2.3 Non-coupling-capable field organization and observability

Within this framework, non-coupling-capable field organization is not to be understood as non-existent or physically ineffective, but merely as not directly visible in the respective observational regime.

Such states nevertheless continue to contribute to the total organization of the field medium and possess gravitational effect. The dark matter effect can therefore be understood as the macroscopic effect of those field components that do not appear in the form of visible, stably bound matter structures, but nonetheless generate persistent deformations and stress states of the field.

In this interpretation, the so-called dark matter halo corresponds not to a particle reservoir, but to a metastable state of large-scale field organization.

5.3 Interpretation within the framework of UQSH

Within the framework of UQSH, the observed regimes can be associated with different coupling and organizational states of the field medium.

UQSH postulates a field equation of the form:

$$\partial_t \phi + v[\phi] \cdot \nabla \phi = \nu \Delta \phi + \gamma \nabla \cdot \left[\sigma \left(\frac{|\nabla \phi|}{S_H} \right) \frac{\nabla \phi}{\sqrt{\varepsilon^2 + |\nabla \phi|^2}} \right] + \kappa R[\phi] - \mu \phi,$$

where:

- $\sigma(z)$ describes an activation function (e.g., $\sigma(z) = z^m/(1 + z^m)$),
- S_H is a characteristic saturation scale,
- $R[\phi]$ represents a nonlocal reorganization term.

5.3.1 Coupling and threshold crossing

Within UQSH, it is assumed that localized reorganization is activated only above a critical threshold:

$$|\nabla \phi| S_{\text{crit}}.$$

Peak regime: in systems where the baryonic structure leads to a local field organization that exceeds this threshold, the reorganization term $R[\phi]$ is activated. This leads to a localized additional field structure with a characteristic length scale.

Diffuse regime: in systems where the threshold is not exceeded, or only weakly so, the reorganization remains large-scale and diffusely distributed.

5.3.2 Scale analogy of localized coupling modes

Within UQSH, the peak regime can be interpreted as the expression of a localized, coupling-capable field mode. This allows a scale-spanning analogy between microscopic and galactic organizational forms.

In this sense, it is conceivable that stable, coupling-capable field modes on the microscopic scale play the role of electrons in a bound system, while on the galactic scale dwarf galaxies and satellite systems appear as analogous bound substructures of a central galactic field node.

The structural correspondence

$$\text{Electrons} \longleftrightarrow \text{Dwarf galaxies}$$

is not to be understood as an identity of the physical forces involved, but as an analogy of organizational form. In both cases, the number of stably bound substructures is determined not solely by their existence, but by their coupling capability to the central field structure.

In this sense, peak systems can be interpreted as localized, coupling-capable field modes, whereas diffuse systems correspond more closely to extended or only partially localized organizational states. This analogy is to be understood as a structural heuristic and does not imply any direct physical equivalence of the underlying interactions. It requires its own independent theoretical elaboration.

5.4 Interpretation of the transition region

The weakly populated but genuinely present transition zone $1.2 < q < 2.5$ suggests that this corresponds to a dynamically less preferred state.

Two physical interpretations are conceivable:

5.4.1 Preferred field modes

Analogous to discrete quantum states, the observed main regimes could correspond to preferred organizational modes of the field medium. In this picture:

- several dynamically preferred field configurations exist,
- the transition region is energetically less favorable,

- systems preferentially relax into one of the more stable modes.

This would be analogous to phase transitions in statistical physics, in which systems preferentially remain in well-defined phases while transition states occur less frequently.

5.4.2 Dynamical transition

Alternatively, the transition region could in principle be accessible, but possess a shorter characteristic lifetime. In this picture:

- systems can pass through the transition region,
- however, the residence time is short compared to the dominant regimes,
- which reduces the probability of observing a system in this state.

From the observed frequency, a rough upper limit for the transition time can be estimated:

$$\tau_{\text{trans}} \frac{n_{\text{trans}}}{n_{\text{total}}} \times \tau_{\text{regime}} \approx 0.12 \times 10 \text{ Gyr} \sim 1.2 \text{ Gyr}.$$

This lies in a range that is compatible with slow dynamical reorganization processes in galactic systems.

5.5 Coupling to baryonic structure

The systematic dependence of the regime structure on the baryonic scaling parameter a indicates that the additional dynamics is not independent of visible matter. This is consistent with the empirically observed close coupling between baryonic and dynamical acceleration, as reflected in the Radial Acceleration Relation [1].

Within UQSH, this means that baryonic structures not only represent local organizational cores of the field, but also act simultaneously as a source of ongoing field excitation and as an anchor of its large-scale stabilization.

In particular, one expects that:

- more strongly concentrated baryonic distributions (high a) lead to large-scale diffuse field response,

- less concentrated distributions (low a) favor the formation of localized reorganization.

This could be related to the fact that strongly concentrated baryonic structures already saturate the local field organization, so that additional reorganization can only occur on large scales.

5.6 Testable predictions

The proposed interpretation leads to several testable predictions:

5.6.1 Galaxy evolution and regime transition

If the observed regimes represent evolutionary stages or different formation pathways, peak and diffuse systems should differ systematically in additional observables:

- gas fraction and star formation rate
- surface brightness
- internal kinematic order
- degree of disturbance

5.6.2 Spatial distribution of subsystems

If peak structures are associated with a more advanced evolutionary state or greater decoupling from the host system, then peak systems should, on average, have larger distances from their host galaxies than diffuse dwarf systems.

This hypothesis can be tested empirically by correlating the projected distance of satellite and dwarf galaxies from their respective host galaxy with the regime classification.

Preliminary analysis. The present evaluation is based on angular separations as an approximation for environmental distance. A more detailed analysis using three-dimensional distances and secure group memberships is the subject of future work.

To investigate a possible environmental dependence of the dynamical regimes identified in the SPARC analysis, an initial association was made between low-mass (dwarf) galaxies and nearby massive host galaxies.

For each dwarf galaxy, a set of plausible host galaxies was constructed from a predefined sample of massive systems. Based on sky coordinates taken from NED, the angular separations between each dwarf galaxy and all potential hosts were calculated. The nearest candidate in each case was taken as an approximation for the dominant external gravitational or field environment. Although this approach does not yet incorporate full three-dimensional distances or dynamical group memberships, it provides a consistent and robust first estimate of environmental coupling.

The resulting distribution of angular separations shows a systematic difference between the previously identified dynamical regimes. Diffuse systems occur across a broad range of separations, including very small angular distances (5°). This suggests strong coupling to nearby massive galaxies.

In contrast, peak systems are not observed at very small separations. Instead, they occur preferentially at intermediate to larger angular distances ($\sim 7^\circ$ – 25°). Both the mean and the median separations are systematically larger for peak systems than for diffuse systems, indicating that a certain degree of decoupling from the immediate galactic environment may be a necessary condition for the formation of a peak structure.

At the same time, the data show that distance alone is not sufficient to determine the dynamical regime uniquely. Systems with comparable angular separations can exhibit either diffuse or peak behavior. This indicates that additional factors – such as internal evolutionary state, baryonic mass distribution, or the organization of the underlying field – play an essential role.

These results support an interpretation in which the effective additional term in rotation curves is not understood as a fixed, universal halo, but as an emergent dynamical quantity whose structure depends both on the baryonic configuration and on the degree of environmental coupling. Within this framework, diffuse systems can be interpreted as strongly coupled or dynamically young states, whereas peak systems represent more strongly decoupled and structurally more evolved configurations.

Although this analysis is preliminary, it provides an initial empirical indication of a correlation between dynamical regime and environmental distance, and motivates further investigations incorporating three-dimensional distances, group catalogs, and kinematic information.

5.6.3 Temporal evolution

The UQSH interpretation implies a causal delay in field reorganization:

$$\tau \sim \frac{L}{c_{\text{eff}}},$$

where L is a characteristic length scale and $c_{\text{eff}} \leq c$ is an effective reorganization velocity.

This leads to the prediction that changes in the baryonic mass distribution do not instantaneously result in a complete adjustment of the non-coupling-capable field structure.

6 Comparison with standard models

6.1 Expectations from the Λ CDM model

Within the framework of the Λ CDM model, the observed additional term is usually interpreted as the contribution of a dark matter halo. The density profiles of such halos (e.g., NFW or Burkert profiles) form a continuous family described by a few parameters.

From this follows the expectation that the resulting rotation curves should exhibit a continuous variety of shapes, without preferred structural states.

In particular, a continuous halo model would predict a more uniform distribution of the scale parameters q across the full range of values.

6.2 Observed discrepancy

The present data show a clear concentration in two preferred regions ($q \approx 0.7$ and $q \approx 3.0$), together with a genuinely present but more weakly populated transition zone.

This structure is not the natural expectation of continuous halo models and requires a further explanation.

The present results extend the coupling between baryonic and dynamical structure known from the Radial Acceleration Relation by adding further information about the shape of the additional dynamics.

6.3 Possible explanation within the Λ CDM framework

In principle, such structural differences could be explained within the standard model by baryonic processes:

- feedback mechanisms (stellar winds, supernova outbursts, AGN activity)
- different formation and evolutionary histories
- morphological differences (spiral galaxies vs. dwarf galaxies)

However, this would require additional assumptions, in particular:

- Why do these processes produce preferred regions in parameter space?
- Why is the transition region systematically less populated?
- Why is there such a strong type dependence within the regimes?

Within the framework of baryonic feedback processes, the observed concentration of systems in preferred parameter regions requires additional assumptions, in particular an explanation for the reduced occupancy of the transition region.

A detailed investigation of these questions within simulations including baryonic physics is left to future work.

6.4 Distinction between the approaches

The proposed field-based interpretation differs from the standard model in several testable respects:

Table 3: Comparison of the predictions of different models for the observed regime structure.

Observable	Λ CDM	UQSH
Number of preferred regimes	continuous	preferred modes
Distribution of q values	smooth	bimodal with transition zone
Dependence on a	weak (indirect)	systematic (direct)
Coupling to baryons	via feedback	fundamental coupling
Spatial correlation	no clear expectation	peak systems farther away
Temporal delay	instantaneous (gravity)	causally limited

7 Discussion and outlook

7.1 Main results

The present work presents a systematic analysis of galactic rotation curves from the SPARC database with the following main results:

1. The empirically extracted additional term $C(r) = v_{\text{obs}}^2 - v_{\text{bar}}^2$ shows, after normalization, a consistent structure of preferred regimes.
2. Global fits across subsamples identify two dominant dynamical regimes:
 - Peak regime ($q \approx 0.5\text{--}1.0$)
 - Diffuse regime ($q \approx 3.0$)
3. Complementary individual fits show a distribution of about 62% peak systems, 12% transition systems, and 26% diffuse systems.
4. The regime structure depends systematically on the baryonic scaling parameter a , indicating a fundamental coupling to the baryonic structure.
5. Within each regime, a universal shape function is found with low scatter ($\text{MSE} \approx 0.016\text{--}0.13$).

The analysis is based on the SPARC database, which constitutes one of the most extensive and homogeneous samples of galactic rotation curves. Verification of the results using independent data sets remains the subject of future work.

7.2 Interpretation and significance

The observed structure with two preferred regions and a more weakly populated transition zone is not the natural expectation of continuous halo models and points to preferred dynamical states.

The present work is to be understood primarily as a data-driven analysis with phenomenological interpretation. A complete theoretical derivation of the observed structures remains to be provided in future work.

Within the field-based interpretation proposed here, these regimes are understood as expressions of different organizational modes of the underlying field medium:

- **Peak regime:** localized reorganization dynamics with a characteristic length scale
- **Diffuse regime:** large-scale field response without local reorganization

The reduced occupancy of the transition region can be interpreted analogously to phase transitions or discrete quantum states, in which preferred configurations exist while transition states are dynamically less stable or energetically less favorable.

7.3 Open questions and future work

Despite the clear empirical structure, several questions remain open and should be the subject of future work:

7.3.1 Physical derivation of the shape function

The parametric form $F(x) = x^p e^{-x/q} / \max(\dots)$ was determined empirically. A quantitative derivation of this form from an underlying field equation is still lacking.

In particular, the following should be clarified:

- Which field equation leads to the observed values $p \approx 1.2$ and $q \approx 0.7$ or $q \approx 3.0$?
- Can these parameters be derived from fundamental field constants?

7.3.2 Scaling relations

The dependence of the parameters $C_{\max}$ and R on the baryonic mass should be investigated systematically. Initial analyses suggest scaling relations of the form

$$C_{\max} \propto M_{\text{bar}}^\alpha, \quad R \propto M_{\text{bar}}^\beta$$

with $\alpha \approx 0.5$ and $\beta \approx 0.3$. A quantitative confirmation of these relations and their physical interpretation remains to be established.

7.3.3 Spatial correlations

The prediction that peak systems should, on average, have larger distances from their host galaxies than diffuse dwarf systems can be tested further empirically. Such an analysis requires:

- identification of host-satellite pairs in the SPARC sample
- determination of projected and, if possible, three-dimensional distances
- statistical correlation analysis with regime classification

7.3.4 Extension to other observables

The analysis so far focuses on rotation curves. An extension to other observables would be desirable:

- Gravitational lensing effects: can the preferred regimes also be identified in lensing signals?
- Velocity dispersion in elliptical galaxies: does an analogous regime structure appear?
- Large-scale structure: can analogous signatures be found in cosmological observations?

7.3.5 Comparison with simulations

A quantitative comparison with numerical simulations is necessary:

- **Λ CDM simulations:** do high-resolution simulations with baryonic physics reproduce the observed bimodal structure?
- **UQSH-based simulations:** can simulations of the proposed field equation quantitatively reproduce the empirical shape functions?

7.4 Significance for the understanding of dark matter

The present results show that the additional dynamics usually attributed to dark matter:

- cannot be described by a completely smooth family of freely adjustable halo profiles,
- but instead assumes preferred structural states,
- which are systematically coupled to the baryonic structure.

This opens the possibility of understanding galactic dynamics as an emergent phenomenon of a deeper, scale-spanning organization rather than attributing it to an additional, freely distributable matter component.

8 Conclusion

In this work, a systematic analysis of 163 galactic rotation curves from the SPARC database was carried out. The empirically extracted additional term $C(r) = v_{\text{obs}}^2 - v_{\text{bar}}^2$ shows, after appropriate normalization, a consistent structure of preferred dynamical regimes.

Global fits across subsamples identify two dominant regimes: a peak regime with $q \approx 0.5$ – 1.0 and a diffuse regime with $q \approx 3.0$.

A complementary analysis of individual fits, however, shows that the overall distribution is broader: approximately 62% of systems lie in the peak regime, about 26% in the diffuse regime, and about 12% in the transition region. The underlying dynamics is therefore not strictly bipartite, but exhibits two preferred states with a genuinely present transition zone.

The systematic dependence of the regime structure on the baryonic scaling parameter shows that the additional dynamics is fundamentally coupled to the baryonic structure.

Within the UQSH framework, this additional dynamics is understood as an expression of persistent, not fully coupling-capable field organization generated and stabilized by baryonic structure and continuous field excitation.

Within the framework of UQSH, the observed regimes are interpreted as expressions of different modes of field organization: the peak regime corresponds to a localized reorganization dynamics, whereas the diffuse regime represents a large-scale field response without local reorganization.

The proposed interpretation leads to testable predictions, in particular regarding the spatial distribution of subsystems, the correlation with further physical properties, and a possible scale-spanning analogy of localized coupling modes.

The present work thus offers an alternative perspective that attributes the observed discrepancy between visible and gravitational mass not to an additional particle component, but to the organization of a continuous field medium.

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References

- [1] McGaugh, S. S., Lelli, F., Schombert, J. M. (2016). Radial Acceleration Relation in Rotationally Supported Galaxies. *Physical Review Letters*, 117, 201101.
- [2] Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *The Astronomical Journal*, 152, 157.
- [3] Rexhepi, U. Q. (2026). Foundations of the Universal Quantum Foam Hypothesis (UQSH): A continuous field medium as the basis of physical reality. Figshare. <https://doi.org/10.6084/m9.figshare.31539715>